

Gradient Structure of the Kuramoto-Type Sequential Associative Memory

Working notes — polynomial DenseNet / Sompolinsky–Kanter synthesis

August 7, 2026

Purpose. This note derives, in full index notation, exactly when the Kuramoto-type phase dynamics proposed for sequential associative memory can be written as gradient descent on a scalar energy $\dot{\theta}_i = -\partial E / \partial \theta_i$, and precisely where that structure breaks down once the Sompolinsky–Kanter asymmetric temporal-association weight and the Krotov polynomial sharpening $f(x) = x^d$ are combined. The single-sequence equation used in the project brief is recovered as one instance of a general $P \times P$ transition-matrix formalism.

1 Setup and notation

Let N be the number of neurons and P the number of stored patterns $\xi^1, \dots, \xi^P \in \{-1, +1\}^N$. Represent each neuron by a phase $\theta_i \in [0, 2\pi)$ with

$$S_i = \cos \theta_i \in [-1, 1], \quad \theta_i \in \{0, \pi\} \Rightarrow S_i \in \{-1, +1\}.$$

Overlap (order parameter). For pattern μ ,

$$m^\mu(\theta) = \frac{1}{N} \sum_{i=1}^N \xi_i^\mu \cos \theta_i, \quad m_i^\mu(\theta) = \frac{1}{N-1} \sum_{j \neq i} \xi_j^\mu \cos \theta_j.$$

m^μ is the global order parameter (used for recall diagnostics); m_i^μ is the leave-one-out version that appears inside the local field acting on neuron i , excluding self-coupling. As $N \rightarrow \infty$ at fixed P the two agree up to $O(1/N)$ and we will not distinguish them except where the distinction matters.

Transition matrix. Collect all pairwise pattern-to-pattern couplings into a single matrix $T \in \mathbb{R}^{P \times P}$, and define the synaptic weight

$$J_{ij} = \frac{1}{N} \sum_{\alpha=1}^P \sum_{\beta=1}^P \xi_i^\alpha T_{\alpha\beta} \xi_j^\beta.$$

Two special cases recover familiar models:

T	Model
$T = I_P$	symmetric auto-associative Hopfield / DAM
$T = \Sigma, \quad \Sigma_{\alpha\beta} = \delta_{\alpha, \beta+1 \bmod P}$	Sompolinsky–Kanter SeqNet (single sequence)
$T = \bigoplus_a \Sigma^{(a)}$	K independent stored sequences

2 The local field: from J_{ij} to m^μ

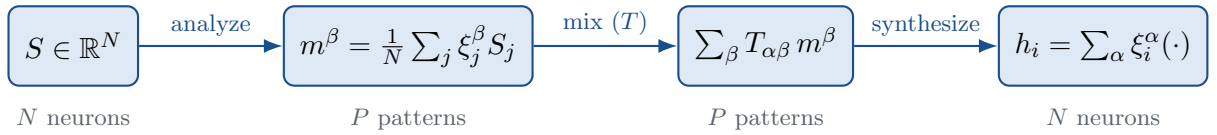
Proposition 1 (Reduction of the local field). *For $h_i := \sum_{j=1}^N J_{ij} S_j$,*

$$h_i = \sum_{\alpha=1}^P \xi_i^\alpha \sum_{\beta=1}^P T_{\alpha\beta} m^\beta(S).$$

Proof. Substitute the definition of J_{ij} and swap the (finite) order of summation, bringing j innermost next to the only other object carrying a j -index:

$$h_i = \sum_{j=1}^N \left(\frac{1}{N} \sum_{\alpha=1}^P \sum_{\beta=1}^P \xi_i^\alpha T_{\alpha\beta} \xi_j^\beta \right) S_j = \sum_{\alpha=1}^P \sum_{\beta=1}^P \xi_i^\alpha T_{\alpha\beta} \left(\frac{1}{N} \sum_{j=1}^N \xi_j^\beta S_j \right) = \sum_{\alpha=1}^P \xi_i^\alpha \sum_{\beta=1}^P T_{\alpha\beta} m^\beta(S). \quad \square$$

Read as a pipeline, this is an *analyze* $\rightarrow$ *mix* $\rightarrow$ *synthesize* factorization:



J_{ij} is what results from collapsing this three-map pipeline into a single fixed $N \times N$ matrix; m^μ is the unavoidable intermediate coordinate the moment the j -sum is carried out on its own. In the Rademacher limit $\Xi^\top \Xi / N \rightarrow I_P$ (crosstalk $O(1/\sqrt{N})$), so $J = \frac{1}{N} \Xi T \Xi^\top$ is precisely T re-expressed in the neuron basis via the (asymptotically) isometric embedding $\Xi/\sqrt{N}$.

3 The symmetric, linear case: a genuine gradient flow

Take $d = 1$ first, so the field of Proposition 1 is the whole local field, and build the natural quadratic energy

$$E(\theta) = -\frac{N}{2} \sum_{\alpha=1}^P \sum_{\beta=1}^P m^\alpha(\theta) T_{\alpha\beta} m^\beta(\theta).$$

Proposition 2 (Gradient of the quadratic energy).

$$-\frac{\partial E}{\partial \theta_i} = -\sin \theta_i \sum_{\alpha=1}^P \xi_i^\alpha \sum_{\beta=1}^P T_{\alpha\beta}^{\text{sym}} m^\beta(\theta), \quad T_{\alpha\beta}^{\text{sym}} := \frac{1}{2}(T_{\alpha\beta} + T_{\beta\alpha}).$$

Proof. Since $\partial m^\gamma / \partial \theta_i = -\frac{1}{N} \xi_i^\gamma \sin \theta_i$ for any pattern index γ , the product rule applied to E gives two terms:

$$\frac{\partial E}{\partial \theta_i} = -\frac{N}{2} \sum_{\alpha,\beta} T_{\alpha\beta} \left[\frac{\partial m^\alpha}{\partial \theta_i} m^\beta + m^\alpha \frac{\partial m^\beta}{\partial \theta_i} \right] = \frac{\sin \theta_i}{2} \left[\sum_{\alpha,\beta} T_{\alpha\beta} \xi_i^\alpha m^\beta + \sum_{\alpha,\beta} T_{\alpha\beta} m^\alpha \xi_i^\beta \right].$$

Relabel the dummy indices $\alpha \leftrightarrow \beta$ in the second sum, so it becomes $\sum_{\alpha,\beta} T_{\beta\alpha} m^\beta \xi_i^\alpha$. Both terms now carry ξ_i^α and m^β , and combine into

$$\frac{\partial E}{\partial \theta_i} = \sin \theta_i \sum_{\alpha} \xi_i^\alpha \sum_{\beta} \frac{1}{2}(T_{\alpha\beta} + T_{\beta\alpha}) m^\beta = \sin \theta_i \sum_{\alpha} \xi_i^\alpha \sum_{\beta} T_{\alpha\beta}^{\text{sym}} m^\beta. \quad \square$$

Only T^{sym} survives differentiation — for any T , without approximation. The antisymmetric part $T^{\text{anti}} = \frac{1}{2}(T - T^\top)$ cancels identically in the relabeling step. This is the exact statement behind the Helmholtz–Hodge intuition: a scalar energy can only ever see the symmetric part of the coupling; whatever drives one-directional sequence traversal must live entirely in T^{anti} and cannot come from any $-\partial E/\partial\theta_i$.

For $T = \Sigma$ (cyclic shift), $\Sigma^\top = \Sigma^{-1}$, so

$$\Sigma^{\text{sym}} = \frac{1}{2}(\Sigma + \Sigma^{-1}), \quad \Sigma^{\text{anti}} = \frac{1}{2}(\Sigma - \Sigma^{-1}).$$

Σ^{sym} couples each pattern symmetrically to *both* neighbors (an elastic-chain stabilizing term); Σ^{anti} is pure circulation, with no potential, and is entirely responsible for one-directional advance through the sequence. The professor’s equation uses Σ alone — i.e. *only* the antisymmetric part, with zero symmetric stabilization, matching why bare SeqNet-type dynamics (linear, $T = \Sigma$, no restoring term) is known to be marginal/unstable and why mixed models reintroduce a symmetric term by hand.

The full, unsymmetrized linear dynamics is therefore not obtained from E at all, but posulated directly from the field of Proposition 1:

$$\dot{\theta}_i = -\sin \theta_i h_i(\theta) = -\sin \theta_i \sum_{\alpha=1}^P \xi_i^\alpha \sum_{\beta=1}^P T_{\alpha\beta} m^\beta(\theta) \quad (d = 1).$$

4 Extending to power d : where the gradient structure breaks

Try the natural generalization of the energy,

$$E(\theta) = -\frac{N}{d+1} \sum_{\alpha,\beta} m^\alpha(\theta) T_{\alpha\beta} (m^\beta(\theta))^d.$$

Proposition 3 (Product-rule mismatch for $d > 1$).

$$\frac{\partial E}{\partial \theta_i} = \sin \theta_i \sum_{\alpha=1}^P \xi_i^\alpha \left[\frac{1}{d+1} \sum_{\beta} T_{\alpha\beta} (m^\beta)^d + \frac{d}{d+1} (m^\alpha)^{d-1} \sum_{\beta} T_{\beta\alpha} m^\beta \right].$$

Proof. Product/chain rule on E , using $\partial m^\gamma/\partial\theta_i = -\frac{1}{N}\xi_i^\gamma \sin \theta_i$ exactly as before:

$$\frac{\partial E}{\partial \theta_i} = \frac{\sin \theta_i}{d+1} \left[\sum_{\alpha,\beta} T_{\alpha\beta} \xi_i^\alpha (m^\beta)^d + d \sum_{\alpha,\beta} T_{\alpha\beta} m^\alpha (m^\beta)^{d-1} \xi_i^\beta \right].$$

Relabel $\alpha \leftrightarrow \beta$ in the second sum ($d \sum_{\alpha,\beta} T_{\beta\alpha} m^\beta (m^\alpha)^{d-1} \xi_i^\alpha$) and factor out ξ_i^α from both terms. $\square$

The two bracketed terms are different functions of m — one is $(m^\beta)^d$, the other is $(m^\alpha)^{d-1} m^\beta$ — and cannot be combined into a single $T_{\alpha\beta}^{\text{sym}} (m^\beta)^d$ for general off-diagonal T . They coincide only at $d = 1$, where $(m^\beta)^1 = m^\beta$ and $(m^\alpha)^0 = 1$ make both terms identical in form; this is exactly why Proposition 2 closes cleanly and Proposition 3 does not.

Remark 1 (Diagonal T is the sole exception). *If $T_{\alpha\beta} = t_\alpha \delta_{\alpha\beta}$, every sum above collapses to $\alpha = \beta$ and the two bracketed terms coincide, giving back a genuine energy*

$$E(\theta) = -\frac{N}{d+1} \sum_{\alpha=1}^P t_\alpha (m^\alpha(\theta))^{d+1}, \quad \dot{\theta}_i = -\sin \theta_i \sum_{\alpha=1}^P \xi_i^\alpha t_\alpha (m^\alpha(\theta))^d.$$

Setting $t_\alpha \equiv 1$ recovers the original single-pattern-set, auto-associative polynomial DAM dynamics.

5 The postulated general dynamics and its coupling tensor

Since Proposition 3 shows no scalar $E(\theta)$ exists for off-diagonal T once $d > 1$, the sequential dynamics is instead built — exactly as the Krotov-DenseNet literature builds it from SeqNet — by inserting the sharpening nonlinearity into m^β inside the linear field of Proposition 1:

$$\dot{\theta}_i = -\sin \theta_i \sum_{\alpha=1}^P \xi_i^\alpha \sum_{\beta=1}^P T_{\alpha\beta} (m^\beta(\theta))^d$$

5.1 Equivalent coupling tensor

Expanding $(m_i^\mu)^d$ into d independent dummy neuron indices,

$$(m_i^\mu)^d = \frac{1}{(N-1)^d} \sum_{j_1 \neq i} \cdots \sum_{j_d \neq i} \xi_{j_1}^\mu \cdots \xi_{j_d}^\mu S_{j_1} \cdots S_{j_d},$$

and substituting into the boxed field shows it is exactly $\sum_{j_1, \dots, j_d} J_{i, j_1, \dots, j_d} S_{j_1} \cdots S_{j_d}$ for the rank- $(d+1)$ tensor

$$J_{i, j_1, \dots, j_d} = \frac{1}{(N-1)^d} \sum_{\alpha=1}^P \sum_{\beta=1}^P \xi_i^\alpha T_{\alpha\beta} \xi_{j_1}^\beta \cdots \xi_{j_d}^\beta,$$

which reduces to J_{ij} at $d = 1$. This tensor is never formed explicitly: because it is built from P rank-one pieces, h_i is evaluated in $O(NP)$ work by routing through $m_i^{(\cdot)}$, rather than $O(N^d)$ by naive tensor contraction.

5.2 Two inequivalent nonlinear extensions

For general T there are two ways to combine mixing and sharpening:

	Field	Coupling tensor
(b) sharpen, then mix	$h_i^{(b)} = \sum_{\alpha, \beta} \xi_i^\alpha T_{\alpha\beta} (m_i^\beta)^d$	$J_{i, j_1, \dots, j_d}^{(b)} = \frac{1}{(N-1)^d} \sum_{\alpha, \beta} \xi_i^\alpha T_{\alpha\beta} \xi_{j_1}^\beta \cdots \xi_{j_d}^\beta$
(a) mix, then sharpen	$h_i^{(a)} = \sum_{\alpha} \xi_i^\alpha \left(\sum_{\beta} T_{\alpha\beta} m_i^\beta \right)^d$	$J_{i, j_1, \dots, j_d}^{(a)} = \frac{1}{(N-1)^d} \sum_{\alpha, \beta_1, \dots, \beta_d} \xi_i^\alpha T_{\alpha\beta_1} \cdots T_{\alpha\beta_d} \xi_{j_1}^{\beta_1} \cdots \xi_{j_d}^{\beta_d}$

Both are exact, fixed tensors (contrary to a naive intractability argument): $J^{(b)}$ forces every one of the d read legs to sample the *same* pattern component β ; $J^{(a)}$ lets each leg carry an *independent* β_k , coupled only through the shared outer index α . They coincide iff T is diagonal. This choice only becomes consequential once off-diagonal T is introduced (skip-transitions, context gating): (b) can only sharpen an already-fixed single overlap, while (a) allows independent nonlinear read-out across several skip-connected patterns before recombination.

6 Reduction to the single-sequence equation

Set $T = \Sigma$, $\Sigma_{\alpha\beta} = \delta_{\alpha,\beta+1 \bmod P}$, in the boxed dynamics. The inner sum collapses to the single term $\beta = \alpha - 1$:

$$\dot{\theta}_i = -\sin \theta_i \sum_{\alpha=1}^P \xi_i^\alpha (m^{\alpha-1})^d.$$

Relabelling $\mu := \alpha - 1$ (periodic, $\xi^{P+1} \equiv \xi^1$) gives $\alpha = \mu + 1$ and

$$\dot{\theta}_i = -\sin \theta_i \sum_{\mu=1}^P \xi_i^{\mu+1} (m^\mu)^d,$$

which is exactly the single-sequence Kuramoto dynamics in the project brief (with $m^\mu \rightarrow m_i^\mu$ as the standard $O(1/N)$ self-interaction correction).

7 Summary

Case	Is $\dot{\theta}_i = -\partial E / \partial \theta_i$ true?	Status
$T = I_P$ (auto-assoc.), any d	Yes	$E = -\frac{N}{d+1} \sum_\mu (m^\mu)^{d+1}$; Krotov–Hopfield DAM.
T symmetric, $d = 1$	Yes	Prop. 2; E quadratic in m .
T symmetric, $d > 1$, off-diagonal	No	Prop. 3: product-rule terms mismatch; only diagonal T salvages a scalar E .
$T = \Sigma$ (Sompolinsky–Kanter), $d = 1$	Only via $T^{\text{sym}} = \frac{1}{2}(\Sigma + \Sigma^{-1})$	Full asymmetric dynamics needs the field postulated directly, not $-\partial E / \partial \theta_i$.
$T = \Sigma$, $d = 3$ (project brief)	No — structurally excluded	Not “not yet found”: Prop. 3 rules it out. Built by analogy, exactly as DenseNet is built from SeqNet in the Krotov literature.

Bottom line. “Derived from the derivative of the energy landscape” is literally correct only for symmetric T (and, for $d > 1$, only if T is further diagonal). The single-sequence equation actually used ($T = \Sigma$, $d = 3$) is not the gradient of any scalar potential: it is constructed by taking the gradient-flow $\sin \theta_i$ structure that *is* rigorously justified in the symmetric case and inserting the Sompolinsky–Kanter asymmetric field in place of the symmetric one — the same by-analogy step the Krotov DenseNet literature uses to go from the linear SeqNet to the polynomial DenseNet update rule. The Helmholtz–Hodge split $T = T^{\text{sym}} + T^{\text{anti}}$ makes this precise at $d = 1$; extending it rigorously to $d > 1$ requires working with the coupling tensor $J_{i,j_1,\dots,j_d}$ directly, since T^{anti} alone is no longer

well-defined as “the non-gradient part” once the tensor legs are degree- d symmetric among themselves but not symmetric with the leading index i .